Xinrui Shi

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Education

- **Shanghai Jiao Tong University (SJTU)** *Sep 2023 – Jun 2027 (Expected)*
  B.S. in Software Engineering, School of Computer Science Shanghai, China
  - GPA: **4.13/4.3**, Ranking: **1/97**
  - Selected Courses: Linear Algebra (A+), Probability and Statistics (A+), Advanced Data Structure (A+), Introduction to Computer Systems (1) (2) (A+), Machine Learning (A+).
- **Stanford University** *Jun 2026 – Aug 2026*
  Undergraduate Visiting Research Intern Stanford, CA, USA

Honors and Awards

- **National Scholarship** (Top 0.4% Nationwide) *2024, 2025, 2026*
- Shanghai Jiao Tong University Undergraduate Category A Scholarship (Top 1% in SJTU) *2024, 2026*
- Shanghai Jiao Tong University Undergraduate Category B Scholarship *2025*
- CCF Elite Collegiate Award (1 of 103 Nationwide) *2026*

Research Interests

My research goal is to build capable and trustworthy AI systems that can operate autonomously in complex real-world environments. Specifically, I'm interested in **agents** and **AI safety**. My broader interests in **multimodal reasoning** and **machine learning systems** focus on capabilities and infrastructure needed for reliable agentic AI.

Publications

[1] **Xinrui Shi**[*], Yanzhe Zhang[*], Diyi Yang. **Emergent Collusion in Long-Horizon LLM Agent Interaction.** Submitted to ICLR 2027. *arXiv preprint arXiv:2609.24967*

[2] **Xinrui Shi**[*], Kai Liu[*], Ziqing Zhang, Jianze Li, Anqi Li, Yulun Zhang. **DRScaffold: Boosting Dense-Scene Reasoning in Lightweight Vision Language Models.** *In Proceedings of the 40th Annual Conference on Neural Information Processing Systems (NeurIPS).* 2026.

[3] Jiaxin Li, Yuxiang Wu, Zhenkai Zhang, **Xinrui Shi**, Haoyuan Wang, Yichen Zhao, Su Linxiang, Chenyang Yu, Mingyu Zhang, Yifan Ding, Boran Wen, Li Zhang, Ruiyang Liu, Yong-Lu Li. **HAT-4D: Lifting Monocular Video for 4D Multi-object Interactions via Human-Agent Collaboration.** *In Proceedings of the 19th European Conference on Computer Vision (ECCV).* 2026.

[4] Tong Xin, **Xinrui Shi**, Mingkai Dong, Zeyu Mi. **Inference in the Shadows: Taming Memory Bandwidth Contention in Mobile LLM Inference with Sereno.** *In Proceedings of the 20th USENIX Symposium on Operating Systems Design and Implementation (OSDI).* 2026.

Research Experiences

- **Multi-Agent Emergent Collusion** 🔗 *May 2026 – Sep 2026*
  SALT lab, Stanford University; Advisor: Diyi Yang Stanford, CA, USA
  - Designed a long-horizon multi-agent framework to study emergent collusion, where agents solve tasks, exchange evidence, verify peers, and adapt from interaction history under conflicting environment.
  - Developed metrics and onset analyses for collusion across 10 frontier LLMs; observed collusion in 93.6% of trajectories with distinct formation patterns across model families.
  - Characterized the dynamics underlying emergent collusion, identifying peer influence, interaction history, and feedback signals as key factors shaping protocol relaxation and behavioral persistence.
  - **Invited talk, Microsoft AI Safety & Security Science Speaker Series, October 2026.**

- **Enhancing Reasoning in Lightweight Vision-Language Models** *Dec 2025 – May 2026*
  Advisor: Yulun Zhang (Shanghai Jiao Tong University) Shanghai, China
  - Designed a dense-scene reasoning benchmark and an automated data generation pipeline with structured annotations to evaluate logical reasoning and hallucination robustness.
  - Developed a structured supervision framework that decomposes dense-scene reasoning into entity grounding, relation modeling, step-wise reasoning, and answer generation.
  - Improved Qwen2.5-VL-3B accuracy by 27.6%, surpassing Qwen2.5-VL-32B while maintaining gains on hallucination benchmarks without degrading general multimodal capabilities.
- **On-device LLM Bandwidth Control** *Aug 2025 – Dec 2025*
  IPADS lab, Shanghai Jiao Tong University; Advisor: Mingkai Dong, Zeyu Mi Shanghai, China
  - Diagnosed memory-bandwidth contention on Snapdragon 8 Elite and Dimensity 9400 platforms, identifying unobservable NPU traffic, missing hardware partitioning support, and queue-depth imbalance.
  - Implemented speculative decoding with fine-grained bandwidth-control mechanisms, and designed a multi-source, length-weighted, self-purifying n-gram cache to recover decoding performance.
  - Evaluated the system on OnePlus 13 against PowerServe, MNN, and llama.cpp across 30 applications, reducing foreground jank rate by 58.5% and improving LLM decode throughput by 26.4% over PowerServe.
- **Monocular 4D Scene Reconstruction via Human-Agent Collaboration** *Dec 2024 – Nov 2025*
  RHOS lab, Shanghai Jiao Tong University; Advisor: Yonglu Li Shanghai, China
  - Developed a multi-level Human-in-the-Loop refinement protocol that enables users' feedback at Gaussian, region, and object levels, and used refined outputs to build a self-improving data engine for 4D reconstruction.
  - Led the construction of a high-fidelity 4D dataset and end-to-end data pipeline, including geometric calibration, temporal synchronization, down-sampling and pixel-level object and key-region annotation via M3-VOS.
  - Introduced a memory metric using DINOv3 for information recovery in occluded regions, and benchmarked against SOTA baselines, improving temporal consistency by 15.2% and occluded-region recovery by 22.4%.

Project Experiences

- **Intelligent Tutoring System** *Jun 2025 – Jul 2025*
  - Built an end-to-end document processing and retrieval pipeline for multi-format course materials, enabling semantic search at both file and page levels.
  - Constructed a course knowledge graph and interactive visualization system to organize and explore relationships among course concepts.
  - Developed a RAG-based tutoring system that integrates vector and knowledge graph retrieval for course question answering.

Skills

Languages: Chinese (Native), English (TOEFL 109; GRE: 162 Verbal / 170 Quant / 4 Analytical Writing)

Programming: C++, Python, Java

Tools: Git, LaTeX, PyTorch, Blender, MySQL, MongoDB, Neo4j